

No. of Functional Features Included in the Solution

Date	01 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Collected loan dataset containing applicant details and loan status.	Data Preprocessing	Done	10%
2	Data Preprocessing	Handled missing values, encoded categorical variables and scaled numerical features.	Data Preprocessing	Done	15%
3	SMOTE Balancing	Applied SMOTE to handle class imbalance in the target variable.	Data Preprocessing	Done	10%
4	Model Building	Trained multiple ML models like Logistic Regression, Decision Tree, Random Forest.	Model Training	Done	15%
5	Model Evaluation	Evaluated models using Accuracy, Precision, Recall, F1-score and Confusion Matrix.	Model Evaluation	Done	15%
6	Best Model Selection	Selected the best performing model (Random Forest) for prediction.	Model Training	Done	5%
7	Prediction Module	Built a module to predict loan approval status for new applicants.	Prediction	Done	10%
8	Web Application (Flask)	Developed a Flask web application with input form and result display.	Deployment (Flask App)	In Progress	10%

Feature Summary:

Metric	Count / Value
Total Features Planned	20
Total Features Implemented	18
Core / Must-Have Features	12
Additional / Nice-to-Have Features	6
Features Tested & Verified	18 / 18 (100%)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Loan application form, prediction result page	Applicant Details Form, Loan Approval/Reject Result
2	Backend / Logic	Data preprocessing, model training, prediction engine	Missing Value Handling, SMOTE, Feature Scaling, Random Forest Prediction
3	Database / Storage	Dataset storage and model files	Loan Dataset (CSV), Trained ML Model (.pkl/.joblib)
4	API / Integration	Flask integration between frontend and ML model	Flask Routes, Prediction API
5	Security / Authentication	Input validation and secure prediction request	Input Validation, Error Handling